

OSSP PRACTICAL Q7:-

7. Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

Code:-

```
#include <stdio.h>
#include <unistd.h>
int main()
{
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child executing ls...\n");
        execl("/bin/ls", "ls", "-l", NULL);
    }
    else
    {
        printf("Parent Waiting...\n");
    }
    return 0;
}
```

OUTPUT:-

```
#include <stdio.h>
#include <unistd.h>
int main()
{
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child executing ls...\n");
        execl("/bin/ls", "ls", "-l", NULL);
    }
    else
```

Parent Waiting...

Child executing ls...

apple@apples-MacBook-Pro-2 OSSP % total 1664

```
-rwxr-xr-x  1 apple  staff  33400 Jul 27 21:23 a.out
-rw-r--r--  1 apple  staff    35 Aug  7 12:29 copy.txt
drwxr-xr-x  6 apple  staff   192 Sep 12 14:54 fifo
drwxr-xr-x  2 apple  staff    64 Aug 17 10:14 fileperm
drwxr-xr-x  4 apple  staff   128 Sep 19 09:07 kernel_monitor
drwxr-xr-x 14 apple  staff   448 Aug 17 13:44 node_modules
-rwxr-xr-x  1 apple  staff  33064 Aug  1 12:20 p1
-rw-r--r--  1 apple  staff    98 Aug  1 12:20 p1.c
-rwxr-xr-x  1 apple  staff  33056 Aug  1 12:23 p2
-rw-r--r--  1 apple  staff    94 Aug  1 12:23 p2.c
-rwxr-xr-x  1 apple  staff  33056 Aug  1 12:25 p3
-rw-r--r--  1 apple  staff   163 Aug  1 12:25 p3.c
-rwxr-xr-x  1 apple  staff  33144 Aug  1 12:27 p4
-rw-r--r--  1 apple  staff   278 Aug  1 12:27 p4.c
-rwxr-xr-x  1 apple  staff  33096 Aug  1 12:29 p5
-rw-r--r--  1 apple  staff   213 Aug  1 12:29 p5.c
-rwxr-xr-x  1 apple  staff  33104 Aug  1 13:39 p6
-rw-r--r--  1 apple  staff   116 Aug  1 13:38 p6.c
-rwxr-xr-x  1 apple  staff  33104 Aug  1 13:40 p7
```